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Linear Algebra

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1 Introduction

2 Vector Space

We consider sets of n scalar numbers $\{x_i \in \mathbf{K} \mid i = 1, 2, \dots, n\}$ and $\{y_i \in \mathbf{K} \mid i = 1, 2, \dots, n\}$ for $\mathbf{K} = \mathbf{R}$ or $\mathbf{K} = \mathbf{C}$, where $\mathbf{R}$ is the set of any real numbers in the open interval $(-\infty, +\infty)$ and $\mathbf{C}$ is the set of any complex numbers except the point at infinity in the complex plain*. From the sets of n scalar numbers, n -dimensional vectors $\mathbf{x}$ and $\mathbf{y}$, and their transpose vectors $\mathbf{x}^T$ $\mathbf{y}^T$ are expressed by means of

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{x}^T = (x_1, x_2, \dots, x_n), \quad (2.1)$$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{y}^T = (y_1, y_2, \dots, y_n). \quad (2.2)$$

For any pair of vectors $\mathbf{x}$ and $\mathbf{y}$,

$$(\mathbf{x}, \mathbf{y}) \equiv \mathbf{x}^T \bar{\mathbf{y}} = \sum_{i=1}^n x_i \bar{y}_i, \quad (2.3)$$

is referred to as an **inner product**, where $\bar{y}_i$ is a complex conjugate of y_i and $\bar{\mathbf{y}} \equiv (\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n)$. For the inner product, we have

$$\overline{(\mathbf{y}, \mathbf{x})} = \overline{\mathbf{y}^T \bar{\mathbf{x}}} = \bar{\mathbf{y}}^T \mathbf{x} = \mathbf{x}^T \bar{\mathbf{y}} = (\mathbf{x}, \mathbf{y}). \quad (2.4)$$

It is easy to confirm that $(\mathbf{x}, \mathbf{x}) \geq 0$ and $(\mathbf{x}, \mathbf{x}) = 0$ is equivalent to $\mathbf{x} = \mathbf{0}$. For any pair of n -dimensional vectors $\mathbf{x}$ and $\mathbf{y}$, if $(\mathbf{x}, \mathbf{y}) = 0$, $\mathbf{x}$ and $\mathbf{y}$ orthogonally intersect of each other. The set of all the n -dimensional vectors is denoted by $\mathbf{K}^n$ and is referred to as a **vector space**. A subset W of the vector space $\mathbf{K}^n$ that satisfies the following three axioms is called a **subspace** of $\mathbf{K}^n$.

Axiom 1 for Subspace: $\mathbf{0} \in W$.

Axiom 2 for Subspace: $\forall \mathbf{x} \in W$ and $\forall \mathbf{y} \in W \Rightarrow \mathbf{x} + \mathbf{y} \in W$.

Axiom 3 for Subspace: $\forall \mathbf{x} \in W$ and $\forall \alpha \in \mathbf{K} \Rightarrow \alpha \mathbf{x} \in W$.

For a set of r vectors in $\mathbf{K}^n$, $\{\mathbf{v}_i \in \mathbf{K}^n \mid i = 1, 2, \dots, r\}$, and a set of scalars in $\mathbf{K}$, $\{c_i \in \mathbf{K} \mid i = 1, 2, \dots, r\}$, the representation

$$\sum_{i=1}^r c_i \mathbf{v}_i = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r, \quad (2.5)$$

is referred to as a **linear combination**. It means that $\{\mathbf{v}_i \in \mathbf{K}^n \mid i = 1, 2, \dots, r\}$ is **linearly dependent**, if it satisfies that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r = \mathbf{0} \Rightarrow \exists i \in \{1, 2, \dots, r\} : c_i \neq 0. \quad (2.6)$$

If $\{\mathbf{v}_i \in \mathbf{K}^n \mid i = 1, 2, \dots, r\}$ is not linearly dependent, we shall say that it is **linearly independent**. It is said that $\{\mathbf{v}_i \in \mathbf{K}^n \mid i = 1, 2, \dots, r\}$ is linearly independent, if it satisfies that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r = \mathbf{0} \Rightarrow c_1 = c_2 = \dots = c_r = 0. \quad (2.7)$$

If there exists n vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in the vector space $\mathbf{K}^n$ which satisfy the following two conditions, the n vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ referred to as **basis** in the vector space $\mathbf{K}^n$:

(a) The n vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent of each other.

(a) Every vector in the vector space $\mathbf{K}^n$ is expressed in terms of a linear combination of the n vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

For a set of n^2 scalar numbers $\{a_{ij} \in \mathbf{K} \mid i = 1, 2, \dots, n, j = 1, 2, \dots, n\}$, $n \times n$ matrix $\mathbf{A}$ and its transpose matrix $\mathbf{A}^T$ are

* In the complex plain, the point at infinity $z = \infty$ is defined so as to satisfy $\frac{1}{z} = 0$.

expressed by means of

$$\begin{aligned} \mathbf{A} &\equiv \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix} = \left(\begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{pmatrix}, \begin{pmatrix} a_{12} \\ a_{22} \\ a_{32} \\ \vdots \\ a_{n2} \end{pmatrix}, \begin{pmatrix} a_{13} \\ a_{23} \\ a_{33} \\ \vdots \\ a_{n3} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ a_{2n} \\ a_{3n} \\ \vdots \\ a_{nn} \end{pmatrix} \right) \\ &= (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_n), \end{aligned} \quad (2.8)$$

and

$$\mathbf{A}^T \equiv \begin{pmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{n1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{n2} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{nn} \end{pmatrix} = \begin{pmatrix} (a_{11}, a_{21}, a_{31}, \dots, a_{n1}) \\ (a_{12}, a_{22}, a_{32}, \dots, a_{n2}) \\ (a_{13}, a_{23}, a_{33}, \dots, a_{n3}) \\ \vdots \\ (a_{1n}, a_{2n}, a_{3n}, \dots, a_{nn}) \end{pmatrix} = \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \\ \vdots \\ \mathbf{a}_n \end{pmatrix}, \quad (2.9)$$

where

$$\mathbf{a}_j \equiv \begin{pmatrix} a_{1j} \\ a_{2j} \\ a_{3j} \\ \vdots \\ a_{nj} \end{pmatrix}, \quad \mathbf{a}_j^T \equiv (a_{1j}, a_{2j}, a_{3j}, \dots, a_{nj}). \quad (2.10)$$

The set of all the matrix in which components belong to $\mathbf{K}^n$ is denoted by $\mathcal{M}_n(\mathbf{K})$.

We introduce the following n vectors:

$$\mathbf{e}_1 \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \mathbf{e}_2 \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \mathbf{e}_3 \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \mathbf{e}_n \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}. \quad (2.11)$$

Every vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ is expressed as

$$\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n \quad (2.12)$$

and it is easy to confirm that

$$x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \dots + x_n \mathbf{e}_n = \mathbf{0} \Rightarrow x_1 = 0, x_2 = 0, \dots, x_n = 0, \quad (2.13)$$

then $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are basis in $\mathbf{K}^n$ and they are referred to as **canonical basis**. We remark that every pair of the basis vectors satisfies the orthonormal relations as follows:

$$(\mathbf{e}_i, \mathbf{e}_j) = \mathbf{e}_i^T \mathbf{e}_j = \delta_{i,j} \quad (i = 1, 2, \dots, n, \quad j = 1, 2, \dots, n). \quad (2.14)$$

By means of the set of the above basis vectors, the representations of the (i, j) -component of the matrix $\mathbf{A}$ are given by

$$a_{ij} = \mathbf{e}_i^T \mathbf{A} \mathbf{e}_j \quad (i = 1, 2, \dots, n, \quad j = 1, 2, \dots, n), \quad (2.15)$$

such that

$$\mathbf{A} = \sum_{k=1}^n \sum_{l=1}^n a_{k,l} \mathbf{e}_k \mathbf{e}_l^T. \quad (2.16)$$

If there exists matrices $\mathbf{X} \in \mathcal{M}_n(\mathbf{K})$ and $\mathbf{Y} \in \mathcal{M}_n(\mathbf{K})$ which satisfy

$$\mathbf{X}\mathbf{A} = \mathbf{I}, \quad \mathbf{A}\mathbf{Y} = \mathbf{I}, \quad (2.17)$$

the matrix A is referred **non-singular matrix**, where I is an $n \times n$ unit matrix which is defined by

$$I \equiv \sum_{k=1}^n \sum_{l=1}^n \delta_{k,l} \mathbf{e}_k \mathbf{e}_l^T = \sum_{l=1}^n \mathbf{e}_l \mathbf{e}_l^T = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}. \quad (2.18)$$

By multiplying X by the both-hand sides of Eq.(2.17), we have

$$(XA)Y = IY = YX(AY) = XI = X, \quad (2.19)$$

and it is valid that $(XA)Y = X(AY)$. It means that $X = Y$. If we have $\tilde{X}$ as another solution of the first equation in Eq.(2.17), such that

$$\tilde{X}A = I, \quad (2.20)$$

Eq.(2.19) is reduced to

$$\tilde{X} = \tilde{X}(AY) = (\tilde{X}A)Y = IY = Y, \quad (2.21)$$

and then is concluded that $X = \tilde{X}$. The non-singularity of the matrix A means that the simultaneous linear equations (2.17) have a unique solution with $X = Y$. Now we denote the solution $X = Y$ by A^{-1} and is referred to as an **inverse matrix** of the matrix A .

For a given $n \times n$ matrix $A = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n) = \sum_{i=1}^n \mathbf{a}_i \mathbf{e}_i$ and a given n -dimensional vector $\mathbf{y} = \sum_{i=1}^n y_i \mathbf{e}_i$, we consider the following simultaneous linear equations for an n -dimensional vector $\mathbf{x}$:

$$A\mathbf{x} = \mathbf{y}. \quad (2.22)$$

Let us suppose that A is a non-singular matrix. By multiplying its inverse matrix A^{-1} by the both-hand sides of Eq.(2.22), we have

$$\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{y}, \quad (2.23)$$

and it means that if A is a non-singular matrix, there exists a set of solution for Eq.(2.22). On the other hand, it is confirmed that if we have $\mathbf{y} = \mathbf{0}$, then Eq.(2.23) have only the trivial solution $\mathbf{x} = \mathbf{0}$, and it means that $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are linearly independent of each other. Moreover, Eq.(2.22) can be reduced to

$$\mathbf{y} = \sum_{i=1}^n x_i \mathbf{a}_i, \quad (2.24)$$

and it means that every vector in the n -dimensional vector space $\mathbf{K}^n$ can be expressed in terms of a linear combination of the set of the n vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$. Hence, for every $n \times n$ non-singular matrix A , the set of n -dimensional vectors, $\{A\mathbf{e}_1, A\mathbf{e}_2, \dots, A\mathbf{e}_n\}$, are basis in the n -dimensional vector space $\mathbf{K}^n$, respectively.

On the other hand, if the set of n -dimensional vectors, $\{A\mathbf{e}_1, A\mathbf{e}_2, \dots, A\mathbf{e}_n\}$, is basis in the n -dimensional vector space $\mathbf{K}^n$, there exists an $n \times n$ matrix $\tilde{A} = \sum_{i=1}^n \sum_{j=1}^n \tilde{a}_{ij} \mathbf{e}_i \mathbf{e}_j^T$ that satisfies the following property:

$$\mathbf{e}_i = \sum_{j=1}^n \tilde{a}_{ji} A\mathbf{e}_j \quad (i = 1, 2, \dots, n). \quad (2.25)$$

Eq.(2.25) can be rewritten to

$$\mathbf{e}_i^T = \sum_{j=1}^n \tilde{a}_{ij} \mathbf{e}_j^T A \quad (i = 1, 2, \dots, n). \quad (2.26)$$

By multiplying Eqs.(2.25) and (2.26) by $\mathbf{e}_i^T$ from the right and by $\mathbf{e}_i^T$ from the left, respectively, and by taking the summations over i , we derive the following equalities:

$$\begin{cases} \sum_{i=1}^n \mathbf{e}_i \mathbf{e}_i^T = \sum_{i=1}^n \sum_{j=1}^n \tilde{a}_{ij} A\mathbf{e}_j \mathbf{e}_i^T = A \left(\sum_{i=1}^n \sum_{j=1}^n \tilde{a}_{ji} \mathbf{e}_j \mathbf{e}_i^T \right), \\ \sum_{i=1}^n \mathbf{e}_i \mathbf{e}_i^T = \sum_{i=1}^n \sum_{j=1}^n \tilde{a}_{ij} \mathbf{e}_i \mathbf{e}_j^T A = \left(\sum_{i=1}^n \sum_{j=1}^n \tilde{a}_{ij} \mathbf{e}_i \mathbf{e}_j^T \right) A. \end{cases} \quad (2.27)$$

Eq.(2.27) means that there exists an $n \times n$ matrix $\tilde{A} = \sum_{i=1}^n \sum_{j=1}^n \tilde{a}_{ij} \mathbf{e}_i \mathbf{e}_j^T$ that satisfying

$$\tilde{A}A = A\tilde{A} = I, \quad (2.28)$$

and then A is a non-singular matrix and $\tilde{A}$ corresponds to its inverse matrix A^{-1} .

It is known that the determinant $\det(A)$ of any $n \times n$ matrix $A = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \mathbf{e}_i \mathbf{e}_j^T$ satisfies the following properties:

$$\det(A) = \det(A^T) \quad (2.29)$$

$$\det\left(\sum_{k=1}^n \sum_{l=1}^n a_{kl} \mathbf{e}_k \mathbf{e}_l^T\right) = -\det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,l} - \delta_{j,l})a_{kl} + \delta_{j,l}a_{ki} + \delta_{i,l}a_{kj}) \mathbf{e}_k \mathbf{e}_l^T\right) \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \\ i \neq j \end{pmatrix}, \quad (2.30)$$

$$\det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,l})a_{kl} + \delta_{i,l}a_{kj}) \mathbf{e}_k \mathbf{e}_l^T\right) = 0 \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \\ i \neq j \end{pmatrix}, \quad (2.31)$$

$$\det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k})a_{kl} + \delta_{i,k}a_{jl}) \mathbf{e}_k \mathbf{e}_l^T\right) = 0 \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \\ i \neq j \end{pmatrix}, \quad (2.32)$$

$$\det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k})a_{kl} + a_{ij}\delta_{i,k}\delta_{j,l}) \mathbf{e}_k \mathbf{e}_l^T\right) = a_{ij} \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k})a_{kl} + \delta_{i,k}\delta_{j,l}) \mathbf{e}_k \mathbf{e}_l^T\right) \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \end{pmatrix}, \quad (2.33)$$

$$\begin{aligned} & \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k} - \delta_{j,l} + \delta_{i,k}\delta_{j,l})a_{kl} + a_{ij}\delta_{i,k}\delta_{j,l}) \mathbf{e}_k \mathbf{e}_l^T\right) \\ &= a_{ij} \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k} - \delta_{j,l} + \delta_{i,k}\delta_{j,l})a_{kl} + \delta_{i,k}\delta_{j,l}) \mathbf{e}_k \mathbf{e}_l^T\right) \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \end{pmatrix}, \end{aligned} \quad (2.34)$$

$$\begin{aligned} \det\left(\sum_{k=1}^n \sum_{l=1}^n a_{kl} \mathbf{e}_k \mathbf{e}_l^T\right) &= \sum_{j=1}^n a_{ij} \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k})a_{kl} + \delta_{i,k}\delta_{j,l}) \mathbf{e}_k \mathbf{e}_l^T\right) \\ &= \sum_{j=1}^n a_{ij} \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{i,k} - \delta_{j,l} + \delta_{i,k}\delta_{j,l})a_{kl} + \delta_{i,k}\delta_{j,l}) \mathbf{e}_k \mathbf{e}_l^T\right) (i = 1, 2, \dots, n). \end{aligned} \quad (2.35)$$

Now we introduce a cofactor of (i, j) -component of the matrix A by the following definition:

$$\text{cof}[A; i, j] \equiv \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{k,i} - \delta_{l,j} + \delta_{i,k}\delta_{j,l})a_{kl} + \delta_{i,j}) \mathbf{e}_k \mathbf{e}_l^T\right) \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \end{pmatrix}. \quad (2.36)$$

From the definition, we derive the following equality:

$$\begin{aligned} \sum_{m=1}^n a_{im} \text{cof}[A; j, m] &= \sum_{m=1}^n a_{im} \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{k,j} - \delta_{l,m} + \delta_{k,j}\delta_{l,m})a_{kl} + \delta_{k,j}\delta_{l,m}) \mathbf{e}_k \mathbf{e}_l^T\right) \\ &= \sum_{m=1}^n \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{k,j} - \delta_{l,m} + \delta_{k,j}\delta_{l,m})a_{kl} + a_{im}\delta_{k,j}\delta_{l,m}) \mathbf{e}_k \mathbf{e}_l^T\right) \\ &= \sum_{m=1}^n \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{k,j})a_{kl} + a_{im}\delta_{k,j}\delta_{l,m}) \mathbf{e}_k \mathbf{e}_l^T\right) \\ &= \det\left(\sum_{k=1}^n \sum_{l=1}^n ((1 - \delta_{k,j})a_{kl} + a_{il}\delta_{k,j}) \mathbf{e}_k \mathbf{e}_l^T\right) \end{aligned}$$

$$\begin{aligned}
&= \delta_{i,j} \det \left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{k,i}) a_{kl} + a_{il} \delta_{k,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \\
&= \delta_{i,j} \det \left(\sum_{k=1}^n \sum_{l=1}^n a_{kl} \mathbf{e}_k \mathbf{e}_l^T \right) = \delta_{i,j} \det(\mathbf{A}) \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \end{pmatrix}, \tag{2.37}
\end{aligned}$$

$$\begin{aligned}
\sum_{m=1}^n a_{mj} \text{cof}[\mathbf{A}; m, i] &= \sum_{m=1}^n a_{mj} \det \left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{k,m} - \delta_{l,i} + \delta_{k,m} \delta_{l,i}) a_{kl} + \delta_{k,m} \delta_{l,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \\
&= \sum_{m=1}^n \det \left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{k,m} - \delta_{l,i} + \delta_{k,m} \delta_{l,i}) a_{kl} + a_{mj} \delta_{k,m} \delta_{l,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \\
&= \sum_{m=1}^n \det \left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{l,i}) a_{kl} + a_{mj} \delta_{k,m} \delta_{l,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \\
&= \det \left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{l,i}) a_{kl} + a_{kj} \delta_{l,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \\
&= \delta_{i,j} \det \left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{l,j}) a_{kl} + a_{kj} \delta_{l,j} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \\
&= \delta_{i,j} \det \left(\sum_{k=1}^n \sum_{l=1}^n a_{kl} \mathbf{e}_k \mathbf{e}_l^T \right) = \delta_{i,j} \det(\mathbf{A}) \begin{pmatrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, n \end{pmatrix}. \tag{2.38}
\end{aligned}$$

Eqs.(2.37) and (2.38) means that we have

$$\mathbf{A} \left(\frac{1}{\det(\mathbf{A})} \sum_{i=1}^n \sum_{j=1}^n \text{cof}[\mathbf{A}; j, i] \mathbf{e}_i \mathbf{e}_j^T \right) = \left(\frac{1}{\det(\mathbf{A})} \sum_{i=1}^n \sum_{j=1}^n \text{cof}[\mathbf{A}; j, i] \mathbf{e}_i \mathbf{e}_j^T \right) \mathbf{A} = \mathbf{I}, \tag{2.39}$$

and then if it is valid that $\det(\mathbf{A}) \neq 0$, $\mathbf{A}$ is a non-singular matrix and there exists an inverse matrix of $\mathbf{A}$ uniquely as follows:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \sum_{i=1}^n \sum_{j=1}^n \text{cof}[\mathbf{A}; j, i] \mathbf{e}_i \mathbf{e}_j^T. \tag{2.40}$$

On the other hand, if $\mathbf{A}$ is a non-singular matrix, there exists an $n \times n$ matrix $\mathbf{X}$ satisfying $\mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{A} = \mathbf{I}$. This equality can be reduced to $\det(\mathbf{A})\det(\mathbf{X}) = \det(\mathbf{X})\det(\mathbf{A}) = 1$ and then $\det(\mathbf{A})$ should be always nonzero.

It is obvious that the solution of inhomogeneous simultaneous linear equations (2.22) is given as Eq.(2.23) with Eq.(2.40) under $\det(\mathbf{A}) \neq 0$. Now we derive another useful formulas to solve inhomogeneous simultaneous linear equations (2.22). By multiplying the both-hand sides of Eq.(2.38) by x_j and by taking the summation with respect to j , we derive

$$\sum_{j=1}^n \left(\sum_{m=1}^n a_{mj} \text{cof}[\mathbf{A}; m, i] \right) x_j = \sum_{j=1}^n (\delta_{i,j} \det(\mathbf{A})) x_j \quad (i = 1, 2, \dots, n). \tag{2.41}$$

$$\sum_{m=1}^n \text{cof}[\mathbf{A}; m, i] \left(\sum_{j=1}^n a_{mj} x_j \right) = \det(\mathbf{A}) \left(\sum_{j=1}^n \delta_{i,j} x_j \right). \quad (i = 1, 2, \dots, n), \tag{2.42}$$

$$\sum_{m=1}^n y_m \text{cof}[\mathbf{A}; m, i] = x_i \det(\mathbf{A}) \quad (i = 1, 2, \dots, n), \tag{2.43}$$

$$\sum_{m=1}^n y_m \det \left(\sum_{k=1}^n \sum_{l=1}^n (1 - \delta_{k,i} - \delta_{l,m} + \delta_{k,i} \delta_{l,m}) a_{kl} + \delta_{k,i} \delta_{l,m} \right) \mathbf{e}_k \mathbf{e}_l^T = x_i \det(\mathbf{A}), \tag{2.44}$$

$$\sum_{m=1}^n \det \left(\sum_{k=1}^n \sum_{l=1}^n (1 - \delta_{k,i} - \delta_{l,m} + \delta_{k,i} \delta_{l,m}) a_{kl} + y_m \delta_{k,i} \delta_{l,m} \right) \mathbf{e}_k \mathbf{e}_l^T = x_i \det(\mathbf{A}) \quad (i = 1, 2, \dots, n), \tag{2.45}$$

$$\sum_{m=1}^n \det \left(\left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{l,i}) a_{kl} + y_k \delta_{l,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \right) = x_i \det(\mathbf{A}) \quad (i = 1, 2, \dots, n). \quad (2.46)$$

$$x_i = \frac{1}{\det(\mathbf{A})} \det \left(\left(\sum_{k=1}^n \sum_{l=1}^n \left((1 - \delta_{l,i}) a_{kl} + y_k \delta_{l,i} \right) \mathbf{e}_k \mathbf{e}_l^T \right) \right) \quad (i = 1, 2, \dots, n). \quad (2.47)$$

Eq.(2.47) is referred to as a **Cramer's rule**.

We consider homogeneous simultaneous linear equations as follows:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = 0, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = 0, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = 0, \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = 0, \end{cases} \quad (2.48)$$

such that

$$\mathbf{A}\mathbf{x} = \mathbf{0}, \quad (2.49)$$

where $a_{ij} = \mathbf{e}_i^T \mathbf{A} \mathbf{e}_j$ and $x_i = \sum_{k=1}^n \mathbf{e}_i^T \mathbf{x}$. If we have $\det(\mathbf{A}) \neq 0$, then there exists its inverse matrix $\mathbf{A}^{-1}$ and we have only the trivial solution $\mathbf{x} = \mathbf{0}$. As this absurd argument, if Eq.(2.49) have not only the trivial solution $\mathbf{x} = \mathbf{0}$ but also non-trivial solutions $\mathbf{x} \neq \mathbf{0}$, it should be valid that $\det(\mathbf{A}) = 0$. On the other hand, if $\det(\mathbf{A}) = 0$, then there does not exist its inverse matrix $\mathbf{A}^{-1}$ and n column vectors $\mathbf{a}_1 = \mathbf{A} \mathbf{e}_1, \mathbf{a}_2 = \mathbf{A} \mathbf{e}_2, \dots, \mathbf{a}_n = \mathbf{A} \mathbf{e}_n$ are not basis and are linearly dependent on each other. It means that Eq.(2.49) has non-trivial solution $\mathbf{x} \neq \mathbf{0}$.

A matrix generated by being extracted r rows $i_1, i_2, \dots, i_r$ ($i_1 < i_2 < \dots < i_r$) and s columns $j_1, j_2, \dots, j_s$ ($j_1 < j_2 < \dots < j_s$) from the $n \times n$ given matrix $\mathbf{A}$ in Eq.(2.8) is expressed as follows:

$$\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_s] = \begin{pmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \dots & a_{i_1, j_s} \\ a_{i_2, j_1} & a_{i_2, j_2} & \dots & a_{i_2, j_s} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \dots & a_{i_r, j_s} \end{pmatrix}, \quad (2.50)$$

and is referred to as a **submatrix** of the matrix $\mathbf{A}$. The maximum number for each size r of submatrix $\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]$ of the given matrix $\mathbf{A}$ satisfying $\det(\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]) \neq 0$ for any sets of integers $\{i_1, i_2, \dots, i_r\} \in \{1, 2, \dots, n\}$ and $\{j_1, j_2, \dots, j_r\} \in \{1, 2, \dots, n\}$ is referred to as a rank of the matrix $\mathbf{A}$ and is denoted by $\text{rank}(\mathbf{A})$.

For every $n \times n$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = n$, Eq.(2.49) has only trivial solution $\mathbf{x} = \mathbf{0}$, because of $\det(\mathbf{A}) \neq 0$. Eq.(2.49) can be reduced to

$$\sum_{i=1}^n x_i \mathbf{A} \mathbf{e}_i = \mathbf{0} \quad (2.51)$$

and then it is concluded that n column vectors $\mathbf{a}_1 = \mathbf{A} \mathbf{e}_1, \mathbf{a}_2 = \mathbf{A} \mathbf{e}_2, \dots, \mathbf{a}_n = \mathbf{A} \mathbf{e}_n$ are linearly independent of each other and are basis. Moreover, the unique solution of inhomogeneous simultaneous linear equations in Eq.(2.22) can be given by

$$\mathbf{x} = \mathbf{A}^{-1} \mathbf{y}. \quad (2.52)$$

In the case of $\text{rank}(\mathbf{A}) < n$, n column vectors $\mathbf{a}_1 = \mathbf{A} \mathbf{e}_1, \mathbf{a}_2 = \mathbf{A} \mathbf{e}_2, \dots, \mathbf{a}_n = \mathbf{A} \mathbf{e}_n$ are not basis and are linearly dependent on each other. For $r = \text{rank}(\mathbf{A})$, there exists the sets of integers $\{i_1, i_2, \dots, i_r\} \in \{1, 2, \dots, n\}$ and $\{j_1, j_2, \dots, j_r\} \in \{1, 2, \dots, n\}$ where $i_1 < i_2 < \dots < i_r$ and $j_1 < j_2 < \dots < j_r$ and $\det(\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]) \neq 0$. In such a case, the homogeneous simultaneous linear equations

$$\begin{cases} a_{i_1 j_1} x_{j_1} + a_{i_1 j_2} x_{j_2} + a_{i_1 j_3} x_{j_3} + \dots + a_{i_1 j_r} x_{j_r} = 0, \\ a_{i_2 j_1} x_{j_1} + a_{i_2 j_2} x_{j_2} + a_{i_2 j_3} x_{j_3} + \dots + a_{i_2 j_r} x_{j_r} = 0, \\ a_{i_3 j_1} x_{j_1} + a_{i_3 j_2} x_{j_2} + a_{i_3 j_3} x_{j_3} + \dots + a_{i_3 j_r} x_{j_r} = 0, \\ \vdots \\ a_{i_r j_1} x_{j_1} + a_{i_r j_2} x_{j_2} + a_{i_r j_3} x_{j_3} + \dots + a_{i_r j_r} x_{j_r} = 0, \end{cases} \quad (2.53)$$

have only the trivial solution $\begin{pmatrix} x_{j_1} \\ x_{j_2} \\ x_{j_3} \\ \vdots \\ x_{j_r} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$. In the present case, Eq.(2.49) can be reduced to the following homogeneous simultaneous linear equations:

$$\begin{cases} a_{i_1 j_1} x_{j_1} + a_{i_1 j_2} x_{j_2} + \cdots + a_{i_1 j_r} x_{j_r} + a_{i_1 j_{r+1}} x_{j_{r+1}} + a_{i_1 j_{r+2}} x_{j_{r+2}} + \cdots + a_{i_1 j_n} x_{j_n} = 0 \\ a_{i_2 j_1} x_{j_1} + a_{i_2 j_2} x_{j_2} + \cdots + a_{i_2 j_r} x_{j_r} + a_{i_2 j_{r+1}} x_{j_{r+1}} + a_{i_2 j_{r+2}} x_{j_{r+2}} + \cdots + a_{i_2 j_n} x_{j_n} = 0 \\ \vdots \\ a_{i_r j_1} x_{j_1} + a_{i_r j_2} x_{j_2} + \cdots + a_{i_r j_r} x_{j_r} + a_{i_r j_{r+1}} x_{j_{r+1}} + a_{i_r j_{r+2}} x_{j_{r+2}} + \cdots + a_{i_r j_n} x_{j_n} = 0 \end{cases} \quad (2.54)$$

Eq.(2.54) can be solved with respect to $x_{j_1}, x_{j_2}, \dots, x_{j_r}$ as follows:

$$\begin{pmatrix} a_{i_1 j_1} & a_{i_1 j_2} & \cdots & a_{i_1 j_r} \\ a_{i_2 j_1} & a_{i_2 j_2} & \cdots & a_{i_2 j_r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r j_1} & a_{i_r j_2} & \cdots & a_{i_r j_r} \end{pmatrix} \begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \vdots \\ x_{j_r} \end{pmatrix} + \begin{pmatrix} a_{i_1 j_{r+1}} & a_{i_1 j_{r+2}} & \cdots & a_{i_1 j_n} \\ a_{i_2 j_{r+1}} & a_{i_2 j_{r+2}} & \cdots & a_{i_2 j_n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r j_{r+1}} & a_{i_r j_{r+2}} & \cdots & a_{i_r j_n} \end{pmatrix} \begin{pmatrix} x_{j_{r+1}} \\ x_{j_{r+2}} \\ \vdots \\ x_{j_n} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad (2.55)$$

$$\begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \vdots \\ x_{j_r} \end{pmatrix} = \begin{pmatrix} a_{i_1 j_1} & a_{i_1 j_2} & \cdots & a_{i_1 j_r} \\ a_{i_2 j_1} & a_{i_2 j_2} & \cdots & a_{i_2 j_r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r j_1} & a_{i_r j_2} & \cdots & a_{i_r j_r} \end{pmatrix}^{-1} \begin{pmatrix} a_{i_1 j_{r+1}} & a_{i_1 j_{r+2}} & \cdots & a_{i_1 j_n} \\ a_{i_2 j_{r+1}} & a_{i_2 j_{r+2}} & \cdots & a_{i_2 j_n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r j_{r+1}} & a_{i_r j_{r+2}} & \cdots & a_{i_r j_n} \end{pmatrix} \begin{pmatrix} x_{j_{r+1}} \\ x_{j_{r+2}} \\ \vdots \\ x_{j_n} \end{pmatrix}. \quad (2.56)$$

All the vectors in the present vector space $\{\mathbf{x} | \mathbf{A}\mathbf{x} = \mathbf{0}\}$ with $\text{rank}(\mathbf{A}) = r < n$ and $\det(\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]) \neq 0$ are expressed as follows:

$$\begin{aligned} \sum_{i=1}^n x_i \mathbf{e}_i &= \sum_{j \in \{j_1, j_2, \dots, j_r\}} \sum_{j' \in \{j_{r+1}, j_{r+2}, \dots, j_n\}} \mathbf{e}_j w_{jj'} x_{j'} + \sum_{j \in \{j_{r+1}, j_{r+2}, \dots, j_n\}} x_j \mathbf{e}_j \\ &= \sum_{j \in \{j_{r+1}, j_{r+2}, \dots, j_n\}} x_j \left(\sum_{j' \in \{j_1, j_2, \dots, j_r\}} w_{jj'} \mathbf{e}_{j'} + \mathbf{e}_j \right), \end{aligned} \quad (2.57)$$

where

$$w_{jj'} \equiv -\mathbf{e}_j^T [j_1, j_2, \dots, j_r] \mathbf{A}^{-1} [j_1, j_2, \dots, j_r; i_1, i_2, \dots, i_r] \mathbf{A} [i_1, i_2, \dots, i_r; j_{r+1}, j_{r+2}, \dots, j_n] \mathbf{e}_{j'} [j_{r+1}, j_{r+2}, \dots, j_n], \quad (j \in \{j_1, j_2, \dots, j_r\}, j' \in \{j_{r+1}, j_{r+2}, \dots, j_n\}), \quad (2.58)$$

$$\begin{aligned} \mathbf{e}_j [j_1, j_2, \dots, j_r] &\equiv \begin{pmatrix} \mathbf{e}_j^T \mathbf{e}_{j_1} \\ \mathbf{e}_j^T \mathbf{e}_{j_2} \\ \vdots \\ \mathbf{e}_j^T \mathbf{e}_{j_r} \end{pmatrix} \quad (j \in \{j_1, j_2, \dots, j_r\}), \\ \mathbf{e}_j [j_{r+1}, j_{r+2}, \dots, j_n] &\equiv \begin{pmatrix} \mathbf{e}_j^T \mathbf{e}_{j_{r+1}} \\ \mathbf{e}_j^T \mathbf{e}_{j_{r+2}} \\ \vdots \\ \mathbf{e}_j^T \mathbf{e}_{j_n} \end{pmatrix} \quad (j \in \{j_{r+1}, j_{r+2}, \dots, j_n\}). \end{aligned} \quad (2.59)$$

The $(n-r)$ vectors $\left(\sum_{j' \in \{j_1, j_2, \dots, j_r\}} w_{jj'} \mathbf{e}_{j'} + \mathbf{e}_j \right)$ ($j \in \{j_{r+1}, j_{r+2}, \dots, j_n\}$) in Eq.(2.57) are linearly independent of each other and then they are basis in the present vector space $\{\mathbf{x} | \mathbf{A}\mathbf{x} = \mathbf{0}\}$ with $\text{rank}(\mathbf{A}) = r < n$ and $\det(\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]) \neq 0$. Instead of the $(n-r)$ vectors $\left(\sum_{j' \in \{j_1, j_2, \dots, j_r\}} w_{jj'} \mathbf{e}_{j'} + \mathbf{e}_j \right)$ ($j \in \{j_{r+1}, j_{r+2}, \dots, j_n\}$) in Eq.(2.57), we can introduce the other set of $n-r$ vectors $\mathbf{x} = \mathbf{x}^{(j_{r+1})}, \mathbf{x}^{(j_{r+2})}, \dots, \mathbf{x}^{(j_n)}$, which satisfies Eq.(2.54) and

are linearly independent of each other, as follows:

$$\mathbf{x}^{(k)} \equiv \sum_{l \in \{j_1, j_2, \dots, j_r\}} x_l^{(k)} \mathbf{e}_l + \mathbf{e}_k \quad (k \in \{j_{r+1}, j_{r+2}, \dots, j_n\}). \quad (2.60)$$

From Eq.(2.60), it is obvious that $n - r$ column vectors $\mathbf{x}^{(k)}$ ($k \in \{j_1, j_2, \dots, j_r\} \setminus \{1, 2, \dots, n\}$) are linearly independent of each other. By substituting Eq.(2.60) to Eq.(??), We have

$$\begin{pmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_r} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_r} \end{pmatrix} \begin{pmatrix} x_{j_1}^{(k)} \\ x_{j_2}^{(k)} \\ \vdots \\ x_{j_r}^{(k)} \end{pmatrix} + \begin{pmatrix} a_{i_1, k} \\ a_{i_2, k} \\ \vdots \\ a_{i_r, k} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (k \in \{j_{r+1}, j_{r+2}, \dots, j_n\}) \quad (2.61)$$

Because of $\det(\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]) \neq 0$, there exists a unique solution of Eq.(2.61) as follows:

$$\begin{pmatrix} x_{j_1}^{(k)} \\ x_{j_2}^{(k)} \\ \vdots \\ x_{j_r}^{(k)} \end{pmatrix} = - \begin{pmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_r} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_r} \end{pmatrix}^{-1} \begin{pmatrix} a_{i_1, k} \\ a_{i_2, k} \\ \vdots \\ a_{i_r, k} \end{pmatrix} \quad (k \in \{j_{r+1}, j_{r+2}, \dots, j_n\}). \quad (2.62)$$

For any $n \times n$ matrix $\mathbf{A}$ with $r = \text{rank}(\mathbf{A})$, there exist $\{i_1, i_2, \dots, i_r\} \in \{1, 2, \dots, n\}$ and $\{j_1, j_2, \dots, j_r\} \in \{1, 2, \dots, n\}$ that satisfy $\det(\mathbf{A}[i_1, i_2, \dots, i_r; 1, 2, \dots, n]) \neq 0$ and there exists a set of $(n - r)$ solutions $\mathbf{x} = \mathbf{x}^{(j_{r+1})}, \mathbf{x}^{(j_{r+2})}, \dots, \mathbf{x}^{(j_n)}$ of $\mathbf{A}[i_1, i_2, \dots, i_r; 1, 2, \dots, n]\mathbf{x} = \mathbf{0}$, which are linearly independent of each other and are basis in the present vector space. Any solution $\mathbf{x}$ of Eq.(2.49), such that any vector $\mathbf{x}$ of the vector space $\{\mathbf{x} | \mathbf{A}[i_1, i_2, \dots, i_r; 1, 2, \dots, n]\mathbf{x} = \mathbf{0}, \mathbf{x} \in \mathbf{K}^n\}$, can be expressed by

$$\mathbf{x} = \sum_{k \in \{j_{r+1}, j_{r+2}, \dots, j_n\}} C_k \mathbf{x}^{(k)}. \quad (2.63)$$

where C_k ($k \in \{j_{r+1}, j_{r+2}, \dots, j_n\}$) are arbitrary constants. For any homogeneous simultaneous linear equations in Eq.(2.49), the set of solutions $\{\mathbf{x}^{(j_{r+1})}, \mathbf{x}^{(j_{r+2})}, \dots, \mathbf{x}^{(j_n)}\}$ is referred to as a **fundamental solution**. Solutions given as Eq.(2.63) is called a **general solution**. On the other hand, one of the solutions ξ in inhomogeneous simultaneous linear equations in Eq.(2.22) for $\text{rank}(\mathbf{A}) = r < n$ with $\det(\mathbf{A}[i_1, i_2, \dots, i_r; j_1, j_2, \dots, j_r]) \neq 0$ can be set to

$$\tilde{\mathbf{x}} = \sum_{l \in \{j_1, j_2, \dots, j_r\}} \tilde{x}_l \mathbf{e}_l. \quad (2.64)$$

By substituting Eq.(2.64) to Eq.(2.22) as $\mathbf{x} = \tilde{\mathbf{x}}$, the simultaneous deterministic equations of $\tilde{\mathbf{x}}$ are reduced to

$$\begin{pmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_r} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_r} \end{pmatrix} \begin{pmatrix} \tilde{x}_{j_1} \\ \tilde{x}_{j_2} \\ \vdots \\ \tilde{x}_{j_r} \end{pmatrix} = \begin{pmatrix} y_{i_1} \\ y_{i_2} \\ \vdots \\ y_{i_r} \end{pmatrix}. \quad (2.65)$$

and then the solution is derived as follows:

$$\begin{pmatrix} \tilde{x}_{j_1} \\ \tilde{x}_{j_2} \\ \vdots \\ \tilde{x}_{j_r} \end{pmatrix} = \begin{pmatrix} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_r} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_r} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_r} \end{pmatrix}^{-1} \begin{pmatrix} y_{i_1} \\ y_{i_2} \\ \vdots \\ y_{i_r} \end{pmatrix}. \quad (2.66)$$

Eq.(2.66) is referred to as a **special solution**. The general solution of Eq.(2.22) is given as

$$\mathbf{x} = \tilde{\mathbf{x}} + \sum_{k \in \{i_{r+1}, i_{r+2}, \dots, i_n\}} C_k \mathbf{x}^{(k)}, \quad (2.67)$$

where $\mathbf{x}^{(k)}$ ($k \in \{i_{r+1}, i_{r+2}, \dots, i_n\}$) and $\tilde{\mathbf{x}}$ are determined from Eqs.(2.62) and (2.66), respectively.

3 Inner Product

We introduce an inner product for any pair of n -dimensional vectors $\mathbf{x}(\in \mathbf{K}^n)$ and $\mathbf{y}(\in \mathbf{K}^n)$ defined by

$$(\mathbf{x}, \mathbf{y}) \equiv \mathbf{x}^T \bar{\mathbf{y}} = \sum_{i=1}^n x_i \bar{y}_i, \quad (3.1)$$

for vector representations in Eqs.(2.1) and (2.2). Inner products have the following properties:

[Propositions for inner products]

- (1) $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbf{K}^n, \forall \alpha, \beta \in \mathbf{K}: (\alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z}) = \alpha(\mathbf{x}, \mathbf{z}) + \beta(\mathbf{y}, \mathbf{z})$
- (2) $\forall \mathbf{x}, \mathbf{y} \in \mathbf{K}^n: (\mathbf{x}, \mathbf{y}) = \overline{(\mathbf{y}, \mathbf{x})} = \overline{(\mathbf{x}, \mathbf{y})}$
- (3) $\forall \mathbf{x} \in \mathbf{K}^n: \mathbf{x} \neq \mathbf{0} \Leftrightarrow (\mathbf{x}, \mathbf{x}) > 0, \mathbf{x} = \mathbf{0} \Leftrightarrow (\mathbf{x}, \mathbf{x}) = 0.$

$$\begin{aligned} (\mathbf{x} + \alpha \mathbf{y}, \mathbf{x} + \alpha \mathbf{y}) &= (\mathbf{x}, \mathbf{x}) + (\mathbf{x}, \alpha \mathbf{y}) + (\alpha \mathbf{y}, \mathbf{x}) + (\alpha \mathbf{y}, \alpha \mathbf{y}) \\ &= (\mathbf{x}, \mathbf{x}) + \overline{\alpha}(\mathbf{x}, \mathbf{y}) + \alpha(\mathbf{y}, \mathbf{x}) + \alpha \overline{\alpha}(\mathbf{y}, \mathbf{y}) \geq 0. \end{aligned} \quad (3.2)$$

By setting $\alpha = -\frac{1}{(\mathbf{y}, \mathbf{y})}(\mathbf{x}, \mathbf{y})$, Eq.(3.2) is reduced to

$$(\mathbf{x}, \mathbf{x}) - \frac{1}{(\mathbf{y}, \mathbf{y})} \overline{(\mathbf{x}, \mathbf{y})}(\mathbf{x}, \mathbf{y}) - \frac{1}{(\mathbf{y}, \mathbf{y})}(\mathbf{x}, \mathbf{y})(\mathbf{y}, \mathbf{x}) + \frac{1}{(\mathbf{y}, \mathbf{y})^2}(\mathbf{x}, \mathbf{y})\overline{(\mathbf{x}, \mathbf{y})}(\mathbf{y}, \mathbf{y}) \geq 0, \quad (3.3)$$

$$(\mathbf{x}, \mathbf{x}) - \frac{1}{(\mathbf{y}, \mathbf{y})} \overline{(\mathbf{x}, \mathbf{y})}(\mathbf{x}, \mathbf{y}) - \frac{1}{(\mathbf{y}, \mathbf{y})}(\mathbf{x}, \mathbf{y})\overline{(\mathbf{x}, \mathbf{y})} + \frac{1}{(\mathbf{y}, \mathbf{y})}(\mathbf{x}, \mathbf{y})\overline{(\mathbf{x}, \mathbf{y})} \geq 0, \quad (3.4)$$

$$(\mathbf{x}, \mathbf{x})(\mathbf{y}, \mathbf{y}) \geq \overline{(\mathbf{x}, \mathbf{y})}(\mathbf{x}, \mathbf{y}). \quad (3.5)$$

The equality is valid either in the case $\mathbf{x} = \mathbf{0}, \mathbf{y} = \mathbf{0}$ or $\mathbf{x} // \mathbf{y}$. The inequality (3.5) is referred to as a **Schwartz inequality**.

For any vector $\mathbf{x}$ in the n -dimensional vector space $\mathbf{K}^n$, a **norm** $\|\mathbf{x}\|$ is defined as something that satisfies the following three axioms:

Axion 1 for Norm: $\forall \mathbf{x} \in \mathbf{K}^n: \|\mathbf{x}\| \geq 0; \|\mathbf{x}\| = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}.$

Axion 2 for Norm: $\forall \mathbf{x}, \mathbf{y} \in \mathbf{K}^n: \|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$

Axion 3 for Norm: $\forall \mathbf{x} \in \mathbf{K}^n, \forall \alpha \in \mathbf{K}: \|\alpha \mathbf{x}\| = |\alpha| \times \|\mathbf{x}\|.$

For any pair of vectors $\mathbf{x}$ and $\mathbf{y}$ in the n -dimensional vector space $\mathbf{K}^n$, $d(\mathbf{x}, \mathbf{y})$ that satisfies the following axioms is called a **distance**:

Axion 1 for Distance: $\forall \mathbf{x}, \mathbf{y} \in \mathbf{K}^n: d(\mathbf{x}, \mathbf{y}) \geq 0; d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}.$

Axion 2 for Distance: $\forall \mathbf{x}, \mathbf{y} \in \mathbf{K}^n: d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x}).$

Axion 3 for Distance: $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbf{K}^n: d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}).$

Let us suppose that $\|\mathbf{x}\| = \sqrt{(\mathbf{x}, \mathbf{x})}$. It is easy to confirm that the above axioms 1 and 3 for norm and we can confirm the axion 3 for the norm as follows:

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 = (\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) &= (\mathbf{x}, \mathbf{x}) + (\mathbf{x}, \mathbf{y}) + (\mathbf{y}, \mathbf{x}) + (\mathbf{y}, \mathbf{y}) \\ &\leq (\mathbf{x}, \mathbf{x}) + 2|(\mathbf{x}, \mathbf{y})| + (\mathbf{y}, \mathbf{y}) \\ &\leq (\mathbf{x}, \mathbf{x}) + 2\sqrt{(\mathbf{x}, \mathbf{x})}\sqrt{(\mathbf{y}, \mathbf{y})} + (\mathbf{y}, \mathbf{y}) \text{ (Schwartz Inequality)} \\ &\leq (\mathbf{x}, \mathbf{x}) + (\mathbf{y}, \mathbf{y}) = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2. \end{aligned} \quad (3.6)$$

We remark that $\|\mathbf{x} - \mathbf{y}\|$ is a distance.

For the n -dimensional vector space $\mathbf{K}^n$, a set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ that satisfy

$$(\mathbf{u}_i, \mathbf{u}_j) = \delta_{i,j} \quad (\forall i, j \in \{1, 2, \dots, m\}), \quad (3.7)$$

is referred to as a **orthonormal system**, where m is a positive integer that is smaller than or equal to n . For the orthonormal system, we consider scalars $x_1, x_2, \dots, x_m$ that satisfy

$$\sum_{i=1}^m x_i \mathbf{u}_i = \mathbf{0}. \quad (3.8)$$

By taking the inner product by $\mathbf{u}_k$ ($k \in \{1, 2, \dots, m\}$) on both hand sides of equation (3.8), respectively, we have

$$\left(\sum_{i=1}^m x_i \mathbf{u}_i, \mathbf{u}_k \right) = (\mathbf{0}, \mathbf{u}_k) \quad (k \in \{1, 2, \dots, m\}), \quad (3.9)$$

$$\sum_{i=1}^m x_i (\mathbf{u}_i, \mathbf{u}_k) = 0 \quad (k \in \{1, 2, \dots, m\}), \quad (3.10)$$

$$\sum_{i=1}^m x_i \delta_{i,k} = 0 \quad (k \in \{1, 2, \dots, m\}), \quad (3.11)$$

$$x_k = 0 \quad (\forall k \in \{1, 2, \dots, m\}). \quad (3.12)$$

It means that the orthonormal system $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is linearly independent.

Let us consider a subspace W , in which all the n -dimensional vectors $\mathbf{x}$ can be expressed in terms of their linear combination of an orthonormal system $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ in the n -dimensional vector space $\mathbf{K}^n$, as follows:

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{u}_i. \quad (3.13)$$

We remark that the subspace W includes $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$. By taking the inner product by $\mathbf{u}_k$ ($k \in \{1, 2, \dots, m\}$) on both hand sides of Eq.(3.13), respectively, we have

$$(\mathbf{x}, \mathbf{u}_k) = \left(\sum_{i=1}^n c_i \mathbf{u}_i, \mathbf{u}_k \right) \quad (\forall k \in \{1, 2, \dots, m\}), \quad (3.14)$$

$$(\mathbf{x}, \mathbf{u}_k) = \sum_{i=1}^n c_i (\mathbf{u}_i, \mathbf{u}_k) \quad (\forall k \in \{1, 2, \dots, m\}), \quad (3.15)$$

$$(\mathbf{x}, \mathbf{u}_k) = \sum_{i=1}^n c_i \delta_{i,k} \quad (\forall k \in \{1, 2, \dots, m\}), \quad (3.16)$$

$$c_k = (\mathbf{x}, \mathbf{u}_k) \quad (\forall k \in \{1, 2, \dots, m\}). \quad (3.17)$$

In this case, it means that every vector $\mathbf{x}$ in the subspace W can be uniquely expressed in terms of Eqs.(3.13) and (3.17) and that the orthonormal system $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is basis of the subspace W . It is said that the the orthonormal system $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is an **orthonormal basis** of the subspace W .

We introduce a set of basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\}$ in a subspace W of the n -dimensional vector space $\mathbf{K}^n$. We set

$$\mathbf{u}_1 \equiv \frac{1}{\|\mathbf{a}_1\|} \mathbf{a}_1, \quad (3.18)$$

and introduce an n -dimensional vector defined by

$$\mathbf{v}_2 \equiv \mathbf{a}_2 + c_{21} \mathbf{u}_1, \quad (3.19)$$

where the coefficient c_{21} is determined so as to satisfy the orthonormal relation $(\mathbf{v}_2, \mathbf{u}_1) = 0$, such that

$$(\mathbf{a}_2 + c_{21} \mathbf{u}_1, \mathbf{u}_1) = 0, \quad (3.20)$$

$$(\mathbf{a}_2, \mathbf{u}_1) + c_{21} (\mathbf{u}_1, \mathbf{u}_1) = 0, \quad (3.21)$$

$$c_{21} = -(\mathbf{a}_2, \mathbf{u}_1). \quad (3.22)$$

Then we set

$$\mathbf{u}_2 \equiv \frac{1}{\|\mathbf{v}_2\|} \mathbf{v}_2 = \frac{\mathbf{a}_2 - (\mathbf{a}_2, \mathbf{u}_1) \mathbf{u}_1}{\|\mathbf{a}_2 - (\mathbf{a}_2, \mathbf{u}_1) \mathbf{u}_1\|} \quad (3.23)$$

Moreover we introduce another n -dimensional vector defined by

$$\mathbf{v}_3 \equiv \mathbf{a}_3 + c_{31} \mathbf{u}_1 + c_{32} \mathbf{u}_2, \quad (3.24)$$

where the coefficient c_{21} is determined so as to satisfy the orthonormal relations $(\mathbf{v}_3, \mathbf{u}_1) = 0$ and $(\mathbf{v}_3, \mathbf{u}_2) = 0$, such that

$$(\mathbf{a}_3 + c_{31}\mathbf{u}_1 + c_{32}\mathbf{u}_2, \mathbf{u}_1) = 0, \quad (\mathbf{a}_3 + c_{31}\mathbf{u}_1 + c_{32}\mathbf{u}_2, \mathbf{u}_2) = 0, \quad (3.25)$$

$$(\mathbf{a}_3, \mathbf{u}_1) + c_{31}(\mathbf{u}_1, \mathbf{u}_1) + c_{32}(\mathbf{u}_2, \mathbf{u}_1) = 0, \quad (\mathbf{a}_3, \mathbf{u}_2) + c_{31}(\mathbf{u}_1, \mathbf{u}_2) + c_{32}(\mathbf{u}_2, \mathbf{u}_2) = 0, \quad (3.26)$$

$$c_{31} = -(\mathbf{a}_3, \mathbf{u}_1), \quad c_{32} = -(\mathbf{a}_3, \mathbf{u}_2). \quad (3.27)$$

Then we set

$$\mathbf{u}_3 \equiv \frac{1}{\|\mathbf{v}_3\|} \mathbf{v}_3 = \frac{\mathbf{a}_3 - (\mathbf{a}_3, \mathbf{u}_1)\mathbf{u}_1 - (\mathbf{a}_3, \mathbf{u}_2)\mathbf{u}_2}{\|\mathbf{a}_3 - (\mathbf{a}_3, \mathbf{u}_1)\mathbf{u}_1 - (\mathbf{a}_3, \mathbf{u}_2)\mathbf{u}_2\|} \quad (3.28)$$

By repeating the similar procedure iteratively, we can generate an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ in the subspace W , such that

$$\begin{cases} \mathbf{u}_1 = \frac{1}{\|\mathbf{a}_1\|} \mathbf{a}_1, \\ \mathbf{u}_k = \frac{\mathbf{a}_k - \sum_{j=1}^{k-1} (\mathbf{a}_k, \mathbf{u}_j) \mathbf{u}_j}{\left\| \mathbf{a}_k - \sum_{j=1}^{k-1} (\mathbf{a}_k, \mathbf{u}_j) \mathbf{u}_j \right\|} \quad (k \in \{2, 3, \dots, m\}). \end{cases} \quad (3.29)$$

The procedure in Eq.(3.29) is referred to as a **Gram-Schmidt orthomormalization**.

Let us $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$ ($\exists r \in \{1, 2, \dots, n\}$) including a subspace W in the n -dimensional vector space is linearly independent and $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r, \mathbf{x}\}$ is always linearly dependent for any n -dimensional vector $\mathbf{x}$ in the subspace $W \setminus \{\mathbf{0}, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$. In such a case, there exists a set $\{c_1, c_2, \dots, c_r\}$ that satisfies

$$\mathbf{x} = \sum_{i=1}^r c_i \mathbf{u}_i, \quad (3.30)$$

where

$$\sum_{i=1}^r |c_i| > 0, \quad (3.31)$$

for any n -dimensional vector $\mathbf{x}$ including the subspace $W \setminus \{\mathbf{0}, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$. It means that such a set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$ is basis in the subspace W . Moreover, we suppose that W is a subspace in which all the vector $\mathbf{x}$ are generated by linear combinations in Eq.(3.30) in terms of the basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$. Any set of vectors $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_s\} \in W \setminus \{\mathbf{0}\}$ ($s > r$) can be expressed as the following linear combination of the basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$:

$$\mathbf{w}_i = \sum_{j=1}^r b_{ji} \mathbf{u}_j \quad (\forall i \in \{1, 2, \dots, s\}), \quad (3.32)$$

where

$$\sum_{i=1}^r |b_{ji}| > 0 \quad (\forall i \in \{1, 2, \dots, s\}). \quad (3.33)$$

If the set of vectors in $W \setminus \{\mathbf{0}, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_s\}$, is linearly independent, it should be valid that

$$\sum_{j=1}^s x_j \mathbf{w}_j = \mathbf{0}, \quad (3.34)$$

only for $x_1 = x_2 = \dots = x_s$. By substituting Eq.(3.32) to Eq.(3.34), we have

$$\sum_{i=1}^r \left(\sum_{j=1}^s b_{ij} x_j \right) \mathbf{u}_i = \mathbf{0}. \quad (3.35)$$

Because the $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$ is linearly independent, we derive the following homogeneous simultaneous linear equations for $x_1, x_2, \dots, x_s$:

$$\sum_{j=1}^s b_{ij} x_j = 0 \quad (i = 1, 2, \dots, r). \quad (3.36)$$

In the case of

$$\text{rank} \begin{pmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,s} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ b_{r,1} & b_{r,2} & \cdots & b_{r,s} \end{pmatrix} = m \leq r, \quad (3.37)$$

there exist two sets of integers, $\{i_1, i_2, \dots, i_m\}$ and $\{j_1, j_2, \dots, j_m\}$ that satisfy $1 \leq i_1 < i_2 < \dots < i_m \leq r$, $1 \leq j_1 < j_2 < \dots < j_m \leq s$ and

$$\det \begin{pmatrix} b_{i_1,j_1} & b_{i_1,j_2} & \cdots & b_{i_1,j_m} \\ b_{i_2,j_1} & b_{i_2,j_2} & \cdots & b_{i_2,j_m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i_m,j_1} & b_{i_m,j_2} & \cdots & b_{i_m,j_m} \end{pmatrix} \neq 0. \quad (3.38)$$

Eq.(3.36) can be reduced to

$$\begin{pmatrix} b_{i_1,j_1} & b_{i_1,j_2} & \cdots & b_{i_1,j_m} \\ b_{i_2,j_1} & b_{i_2,j_2} & \cdots & b_{i_2,j_m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i_m,j_1} & b_{i_m,j_2} & \cdots & b_{i_m,j_m} \end{pmatrix} \begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \vdots \\ x_{j_m} \end{pmatrix} + \begin{pmatrix} b_{i_1,j_{m+1}} & b_{i_1,j_{m+2}} & \cdots & b_{i_1,j_s} \\ b_{i_2,j_{m+1}} & b_{i_2,j_{m+2}} & \cdots & b_{i_2,j_s} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i_m,j_{m+1}} & b_{i_m,j_{m+2}} & \cdots & b_{i_m,j_s} \end{pmatrix} \begin{pmatrix} x_{j_{m+1}} \\ x_{j_{m+2}} \\ \vdots \\ x_{j_s} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad (3.39)$$

and it is concluded that there exist non-trivial solutions of Eq.(3.39) as follows:

$$\begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \vdots \\ x_{j_m} \end{pmatrix} = - \begin{pmatrix} b_{i_1,j_1} & b_{i_1,j_2} & \cdots & b_{i_1,j_m} \\ b_{i_2,j_1} & b_{i_2,j_2} & \cdots & b_{i_2,j_m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i_m,j_1} & b_{i_m,j_2} & \cdots & b_{i_m,j_m} \end{pmatrix}^{-1} \begin{pmatrix} b_{i_1,j_{m+1}} & b_{i_1,j_{m+2}} & \cdots & b_{i_1,j_s} \\ b_{i_2,j_{m+1}} & b_{i_2,j_{m+2}} & \cdots & b_{i_2,j_s} \\ \vdots & \vdots & \ddots & \vdots \\ b_{i_m,j_{m+1}} & b_{i_m,j_{m+2}} & \cdots & b_{i_m,j_s} \end{pmatrix} \begin{pmatrix} x_{j_{m+1}} \\ x_{j_{m+2}} \\ \vdots \\ x_{j_s} \end{pmatrix}. \quad (3.40)$$

It means that the set of vectors in $W \setminus \{\mathbf{0}\}$, $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_s\}$, is linearly dependent for $r < s$. Moreover, if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$ and $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_s\}$ are bases in the subspace W , it should be valid that $r = s$.

If there exists some pairs of λ and $\mathbf{u}$ that satisfy

$$\exists \lambda \in \mathbf{K}, \exists \mathbf{u} \in \mathbf{K}^n \setminus \{\mathbf{0}\}; \mathbf{A}\mathbf{u} = \lambda\mathbf{u}, \quad (3.41)$$

for a $n \times n$ matrix $\mathbf{A}$, λ and its corresponding $\mathbf{u}$ are referred to as an **eigenvalue** and an **eigenvector** of the matrix $\mathbf{A}$, respectively. Eq.(3.41) can be reduced to

$$(\mathbf{A} - \lambda \mathbf{I})\mathbf{u} = \mathbf{0}. \quad (3.42)$$

The necessary and sufficient condition that the homogeneous simultaneous linear equations (3.42) with respect to $\mathbf{u}$ have non-trivial solutions is given as

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0. \quad (3.43)$$

where Eq.(3.43) is referred to as a **characteristic equation**. For any pair of vectors, $\mathbf{u}_1$ and $\mathbf{u}_2$, that satisfy $\mathbf{A}\mathbf{u}_1 = \lambda\mathbf{u}_1$ and $\mathbf{A}\mathbf{u}_2 = \lambda\mathbf{u}_2$ for a fixed value $\lambda \in \mathbf{K}$, we have $\mathbf{A}(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) = \lambda(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2)$ for any values $\alpha \in \mathbf{K}$ and $\beta \in \mathbf{K}$. It means that if $\mathbf{u}_1$ and $\mathbf{u}_2$ belong to $\{\mathbf{u} \in \mathbf{K}^n | \mathbf{A}\mathbf{u} = \lambda\mathbf{u}\}$, $\alpha\mathbf{u}_1 + \beta\mathbf{u}_2$ is always included to $\{\mathbf{u} \in \mathbf{K}^n | \mathbf{A}\mathbf{u} = \lambda\mathbf{u}\}$ and then $\{\mathbf{u} \in \mathbf{K}^n | \mathbf{A}\mathbf{u} = \lambda\mathbf{u}\}$ is a subspace of the n -dimensional vector space $\mathbf{K}^n$.

Let us consider an $n \times n$ matrix $\mathbf{U}$, that the set of n -dimensional column vectors $\{\mathbf{U}\mathbf{e}_1, \mathbf{U}\mathbf{e}_2, \dots, \mathbf{U}\mathbf{e}_n\}$ is a basis in the n -dimensional vector space $\mathbf{K}^n$, such that $\det(\mathbf{A}) \neq 0$ and there exists $\mathbf{A}^{-1}$. Two $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ that satisfy $\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{B}$, such that $\mathbf{B} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$ are said **similar** to each other and we have

$$\det(\mathbf{B} - \lambda \mathbf{I}) = \det(\mathbf{U}^{-1}\mathbf{A}\mathbf{U} - \lambda \mathbf{I}) = \det(\mathbf{U}^{-1}(\mathbf{A} - \lambda \mathbf{I})\mathbf{U}) = \det(\mathbf{U}^{-1})\det(\mathbf{A} - \lambda \mathbf{I})\det(\mathbf{U}) = \det(\mathbf{A} - \lambda \mathbf{I}). \quad (3.44)$$

It means that the characteristic equations of mutually similar matrices are equal to each other. For a $n \times n$ matrix $\mathbf{A}$, if there exists a matrix $\mathbf{U}$ that satisfy

$$\mathbf{U}^{-1}\mathbf{A}\mathbf{U} = \sum_{i=1}^n \lambda_i \mathbf{e}_i \mathbf{e}_i^T = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_{n-1} & 0 \\ 0 & 0 & 0 & \cdots & 0 & \lambda_n \end{pmatrix}, \quad (3.45)$$

it is said that A is **diagonalizable**. Eq.(3.45) can be reduced as follows:

$$AU \left(\sum_{i=1}^n \mathbf{e}_i \mathbf{e}_i^T \right) = U \left(\sum_{i=1}^n \lambda_i \mathbf{e}_i \mathbf{e}_i^T \right) \quad (3.46)$$

$$\sum_{i=1}^n (AU \mathbf{e}_i) \mathbf{e}_i^T = \sum_{i=1}^n (\lambda_i U \mathbf{e}_i) \mathbf{e}_i^T \quad (3.47)$$

$$\left(\sum_{i=1}^n (AU \mathbf{e}_i) \mathbf{e}_i^T \right) \mathbf{e}_k = \left(\sum_{i=1}^n (\lambda_i U \mathbf{e}_i) \mathbf{e}_i^T \right) \mathbf{e}_k \quad (3.48)$$

$$\sum_{i=1}^n (AU \mathbf{e}_i) (\mathbf{e}_i^T \mathbf{e}_k) = \sum_{i=1}^n (\lambda_i U \mathbf{e}_i) (\mathbf{e}_i^T \mathbf{e}_k) \quad (3.49)$$

$$\sum_{i=1}^n (AU \mathbf{e}_i) \delta_{i,k} = \sum_{i=1}^n (\lambda_i U \mathbf{e}_i) \delta_{i,k} \quad (3.50)$$

$$AU \mathbf{e}_k = \lambda_k U \mathbf{e}_k \quad (k \in \{1, 2, \dots, n\}). \quad (3.51)$$

Eq.(3.51) is equivalent to Eq.(3.45). The necessary and sufficient condition that the matrix A is diagonalizable is that A has n eigenvectors, $\mathbf{u}_1 = U \mathbf{e}_1, \mathbf{u}_2 = U \mathbf{e}_2, \dots, \mathbf{u}_n = U \mathbf{e}_n$, which are linearly independent of each other.

By introducing an $n \times n$ matrix V_1 satisfying

$$\begin{cases} \bar{\mathbf{V}}_1^T V_1 = V_1 \bar{\mathbf{V}}_1^T = I, \\ \mathbf{e}_i^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_1 = \lambda_1 \delta_{i,1} \quad (i = 1, 2, \dots, n). \end{cases} \quad (3.52)$$

we can derive the following equality:

$$\begin{aligned} \bar{\mathbf{V}}_1^T A V_1 &= \begin{pmatrix} \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_1 & \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \\ \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_1 & \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \\ \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_1 & \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_1 & \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \end{pmatrix} \\ &= \begin{pmatrix} \lambda_1 & \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_1^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \\ 0 & \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_2^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \\ 0 & \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_3^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_2 & \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_3 & \cdots & \mathbf{e}_n^T (\bar{\mathbf{V}}_1^T A V_1) \mathbf{e}_n \end{pmatrix}. \end{aligned} \quad (3.53)$$

Next step, we choose an $n \times n$ matrix V_2 satisfying

$$\begin{cases} \bar{\mathbf{V}}_2^T V_2 = V_2 \bar{\mathbf{V}}_2^T = I, \\ \mathbf{e}_i^T V_2 \mathbf{e}_1 = \delta_{i,1}, \quad \mathbf{e}_1^T V_2 \mathbf{e}_i = \delta_{1,i} \quad (i = 1, 2, \dots, n), \\ \mathbf{e}_i^T (\bar{\mathbf{V}}_2^T (\bar{\mathbf{V}}_1^T A V_1) V_2) \mathbf{e}_2 = \lambda_2 \delta_{i,2} \quad (i = 2, 3, 4, \dots, n). \end{cases} \quad (3.54)$$

we obtain

$$\begin{aligned}
\bar{V}_2^T (\bar{V}_1^T A V_1) V_2 &= \bar{V}_2^T \begin{pmatrix} \lambda_1 & e_1^T (\bar{V}_1^T A V_1) e_2 & e_1^T (\bar{V}_1^T A V_1) e_3 & \cdots & e_1^T (\bar{V}_1^T A V_1) e_n \\ 0 & e_2^T (\bar{V}_1^T A V_1) e_2 & e_2^T (\bar{V}_1^T A V_1) e_3 & \cdots & e_2^T (\bar{V}_1^T A V_1) e_n \\ 0 & e_3^T (\bar{V}_1^T A V_1) e_2 & e_3^T (\bar{V}_1^T A V_1) e_3 & \cdots & e_3^T (\bar{V}_1^T A V_1) e_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & e_n^T (\bar{V}_1^T A V_1) e_2 & e_n^T (\bar{V}_1^T A V_1) e_3 & \cdots & e_n^T (\bar{V}_1^T A V_1) e_n \end{pmatrix} V_2 \\
&= \begin{pmatrix} \lambda_1 & e_1^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_2 & e_1^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_1^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \\ 0 & e_2^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_2 & e_2^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_2^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \\ 0 & e_3^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_2 & e_3^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_3^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & e_n^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_2 & e_n^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_n^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \end{pmatrix} \\
&= \begin{pmatrix} \lambda_1 & e_1^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_2 & e_1^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_1^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \\ 0 & \lambda_2 & e_2^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_2^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \\ 0 & 0 & e_3^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_3^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & e_n^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_3 & \cdots & e_n^T (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) e_n \end{pmatrix} \quad (3.55)
\end{aligned}$$

By doing the similar procedures iteratively, we derive the following equality:

$$\begin{aligned}
\bar{U}^T A U &= (\bar{V}_n^T \cdots (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) \cdots V_n) \\
&= \begin{pmatrix} \lambda_1 & e_1^T (\bar{U}^T A U) e_2 & e_1^T (\bar{U}^T A U) e_3 & \cdots & e_1^T (\bar{U}^T A U) e_n \\ 0 & \lambda_2 & e_2^T (\bar{U}^T A U) e_3 & \cdots & e_2^T (\bar{U}^T A U) e_n \\ 0 & 0 & \lambda_3 & \cdots & e_3^T (\bar{U}^T A U) e_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{pmatrix}, \quad (3.56)
\end{aligned}$$

where

$$U \equiv V_1 V_2 V_3 \cdots V_n, \quad (3.57)$$

$$\begin{cases} \bar{V}_k^T V_k = V_k \bar{V}_k^T = I \ (k = 1, 2, \dots, n), \\ e_i^T V_k e_j = \delta_{i,j}, \ e_j^T V_k e_i = \delta_{j,i} \ (i = 1, 2, \dots, n, \ j = 1, 2, \dots, k-1), \\ e_i^T (\bar{V}_k^T \cdots (\bar{V}_2^T (\bar{V}_1^T A V_1) V_2) \cdots V_k) e_k = \lambda_k \delta_{i,k} \ (i = 2, 3, 4, \dots, n). \end{cases} \quad (3.58)$$

Moreover, if the $n \times n$ matrix A is an **Hermite matrix** satisfying

$$\bar{A}^T A = A \bar{A}^T = I, \quad (3.59)$$

such that

$$A^{-1} = \bar{A}^T. \quad (3.60)$$

It is easily confirmed that the above matrix U in Eq.(3.57) is a **unitary matrix** satisfying

$$\bar{U}^T U = U \bar{U}^T = I, \quad (3.61)$$

and then it is valid that

$$\bar{U}^T A U (\overline{\bar{U}^T A U})^T = \bar{U}^T A U \bar{U}^T \bar{A}^T U = \bar{U}^T A \bar{A}^T U = \bar{U}^T \bar{A}^T A U = \bar{U}^T \bar{A}^T U \bar{U}^T A U = (\overline{\bar{U}^T A U})^T \bar{U}^T A U. \quad (3.62)$$

Moreover we have

$$\mathbf{e}_i^T \bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} \left(\overline{\bar{\mathbf{U}}^T \mathbf{A} \mathbf{U}} \right)^T \mathbf{e}_i = \mathbf{e}_i^T \left(\overline{\bar{\mathbf{U}}^T \mathbf{A} \mathbf{U}} \right)^T \bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} \mathbf{e}_i \quad (i = 1, 2, \dots, n) \quad (3.63)$$

$$\lambda_i \bar{\lambda}_i + \sum_{j=i+1}^n \mathbf{e}_i^T \bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} \mathbf{e}_j \mathbf{e}_j^T \left(\overline{\bar{\mathbf{U}}^T \mathbf{A} \mathbf{U}} \right)^T \mathbf{e}_i = \lambda_i \bar{\lambda}_i \quad (i = 1, 2, \dots, n), \quad (3.64)$$

$$\sum_{j=i+1}^n \mathbf{e}_i^T \bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} \mathbf{e}_j \left(\overline{\mathbf{e}_i^T \bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} \mathbf{e}_j} \right) = 0 \quad (i = 1, 2, \dots, n), \quad (3.65)$$

and then it is valid that

$$\mathbf{e}_i^T \bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} \mathbf{e}_j = 0 \quad (i = 1, 2, \dots, n, \quad j = i+1, i+2, \dots, n). \quad (3.66)$$

Eq.(3.66) concludes that the Hermite matrix $\mathbf{A}$ is diagonalizable as follows:

$$\bar{\mathbf{U}}^T \mathbf{A} \mathbf{U} = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{pmatrix}, \quad (3.67)$$

and it means that, for any Hermite matrix $\mathbf{A}$, we have found a unitary matrix $\mathbf{U}$ satisfying Eq.(3.67) by defined in Eqs.(3.57) and (3.58).

For any pair of an Hermite matrix $\mathbf{A}$ and a unitary matrix $\mathbf{U}$ satisfying Eq.(3.67), n vectors

$$\begin{cases} \mathbf{u}_1 = \mathbf{U} \mathbf{e}_1, \\ \mathbf{u}_2 = \mathbf{U} \mathbf{e}_2, \\ \vdots \\ \mathbf{u}_n = \mathbf{U} \mathbf{e}_n, \end{cases} \quad (3.68)$$

satisfy

$$\begin{cases} \mathbf{A} \mathbf{u}_i = \lambda_i \mathbf{u}_i, \\ \bar{\mathbf{u}}_i^T \mathbf{A}^T = \bar{\lambda}_i \bar{\mathbf{u}}_i^T \quad \left(\overline{\mathbf{A} \mathbf{u}_i}^T = \bar{\lambda}_i \bar{\mathbf{u}}_i^T \Rightarrow \bar{\mathbf{u}}_i^T \mathbf{A}^T = \bar{\mathbf{u}}_i^T \mathbf{A} = \bar{\lambda}_i \bar{\mathbf{u}}_i^T \right), \end{cases} \quad (i = 1, 2, \dots, n). \quad (3.69)$$

From Eq.(3.68) we can confirm that

$$\left(\overline{\mathbf{A} \mathbf{u}_i} \right)^T \mathbf{u}_j = \bar{\mathbf{u}}_i^T \mathbf{A}^T \mathbf{u}_j = \bar{\mathbf{u}}_i^T \mathbf{A} \mathbf{u}_j = \bar{\mathbf{u}}_i^T (\mathbf{A} \mathbf{u}_j) \quad (i = 1, 2, \dots, n \quad j = 1, 2, \dots, n), \quad (3.70)$$

and

$$\begin{cases} \left(\overline{\mathbf{A} \mathbf{u}_i} \right)^T \mathbf{u}_j = \bar{\mathbf{u}}_i^T \mathbf{A} \mathbf{u}_j = \bar{\lambda}_i \bar{\mathbf{u}}_i^T \mathbf{u}_j, \\ \bar{\mathbf{u}}_i^T (\mathbf{A} \mathbf{u}_j) = \bar{\mathbf{u}}_i^T (\lambda_j \mathbf{u}_j) = \lambda_j \bar{\mathbf{u}}_i^T \mathbf{u}_j, \end{cases} \quad (i = 1, 2, \dots, n \quad j = 1, 2, \dots, n). \quad (3.71)$$

From Eqs.(3.69) and (3.69), we obtain

$$(\lambda_i - \bar{\lambda}_i) \bar{\mathbf{u}}_i^T \mathbf{u}_i = 0 \Rightarrow \lambda_i = \bar{\lambda}_i \quad (i = 1, 2, \dots, n), \quad (3.72)$$

and

$$\lambda_i \neq \lambda_j \Rightarrow (\lambda_i - \bar{\lambda}_j) \bar{\mathbf{u}}_i^T \mathbf{u}_j = (\lambda_i - \lambda_j) \bar{\mathbf{u}}_i^T \mathbf{u}_j = 0 \Rightarrow \bar{\mathbf{u}}_i^T \mathbf{u}_j = 0 \quad (i = 1, 2, \dots, n, \quad j = 1, 2, \dots, n, \quad i \neq j). \quad (3.73)$$

Eq.(3.72) means that all the eigenvalues of any Hermite matrix $\mathbf{A}$ are always real numbers. Eq.(3.73) means that if all the eigenvalues of any Hermite matrix $\mathbf{A}$ are not degenerate of each other at all, all the corresponding eigenvectors can be chosen as mutual linearly independent vectors, such that the corresponding unitary matrix $\mathbf{U}$ can be also chosen as an Hermite matrix satisfying

$$\begin{cases} \mathbf{U} \bar{\mathbf{U}}^T = \bar{\mathbf{U}}^T \mathbf{U} = \mathbf{I}, \\ \mathbf{U}^{-1} = \bar{\mathbf{U}}^T. \end{cases} \quad (3.74)$$

For any Hermite matrix A , we introduce the following so-called **Hermite form**:

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} \bar{x}_i x_j = \bar{\mathbf{x}}^T A \mathbf{x} = \sum_{i=1}^n \bar{\mathbf{u}}^T \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{pmatrix} \mathbf{u} \quad (3.75)$$

Because we have

$$\overline{\bar{\mathbf{x}}^T A \mathbf{x}} = \mathbf{x}^T \overline{A \mathbf{x}} = \sum_{i=1}^n \sum_{j=1}^n \bar{a}_{ij} x_i \bar{x}_j = \sum_{i=1}^n \sum_{j=1}^n a_{ji} \bar{x}_j x_i = \bar{\mathbf{x}}^T A \mathbf{x}, \quad (3.76)$$

$\overline{\bar{\mathbf{x}}^T A \mathbf{x}}$ is always real number. If $\bar{\mathbf{x}}^T A \mathbf{x}$ is always positive for any vector $\mathbf{x} \in \mathbf{K}^n$, the Hermite matrix A is said **potitive definite** and all the eigenvalues are always positive. Moreover, if $\bar{\mathbf{x}}^T A \mathbf{x}$ is always non-negative for any vector $\mathbf{x} \in \mathbf{K}^n$, the Hermite matrix A is said **potitive semi-definite** and all the eigenvalues are always non-negative.

If there exist a set of positive integers $\{m_0, m_1, m_2, \dots, m_L | 0 = m_0 < m_1 < m_2 < \dots < m_L = n\}$ satisfying

$$\det(A - \lambda I) = \prod_{l=1}^L (\lambda - \lambda_{m_l})^{m_l - m_{l-1}}, \quad (3.77)$$

we have

$$\bar{\mathbf{U}}^T A \mathbf{U} = \sum_{l=1}^L \lambda_{m_l} \left(\sum_{j=m_l}^{m_{l+1}-1} \mathbf{e}_j^T \mathbf{e}_j \right). \quad (3.78)$$

It means that we have $m_l - m_{l-1}$ mutual linearly independent eigenvectors $\mathbf{u}_{m_{l-1}+1}, \mathbf{u}_{m_{l-1}+2}, \dots, \mathbf{u}_{m_l}$ satisfying

$$A \mathbf{u}_j = \lambda_{m_l} \mathbf{u}_j \quad (j = m_{l-1} + 1, m_{l-1} + 2, \dots, m_l, l = 1, 2, \dots, L). \quad (3.79)$$

References

- [1] F. Uchida and H. Urakawa: *Linear Algebra*, Shokabo, 2000 (ISBN978-4-7853-1522-1).
- [2] I. Satake: *Linear Algebra*, Shokabo, 1997 (ISBN978-4-320-01554-8).
- [3] S. Lang: *Linear Algebra*, Springer, 1987 (ISBN978-1-4419-308101).
- [4] D. A. Harville: *Matrix Algebra from a Statisticain's Perspective*, Springer, 1997 (ISBN0-387-94978-X).